

AYMANE AIT DADS

AI/ML Engineering Intern | Deep Learning · Signal Processing · Model Benchmarking

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PROFESSIONAL SUMMARY

Engineering student (Master-level, EURECOM) with hands-on deep learning training and optimization experience across supervised, unsupervised, and generative architectures, including transformer autoencoders, ViT, and Mamba-Transformer hybrid models on real-world datasets exceeding 100K records. Built and deployed an end-to-end ML pipeline at Orange Maroc processing 100K+ fiber-optic network records, reducing manual analysis time by ~60% through automated feature engineering and ensemble clustering - directly demonstrating reproducible benchmarking and comparative model evaluation in a production context. Proficient in Python, PyTorch, Scikit-learn, and Hugging Face Transformers, with additional experience in state-space models, LoRA fine-tuning, and ablation-driven architecture comparison across modern deep learning frameworks. Available for a 3-6 month internship starting immediately; located in Sophia Antipolis, France, with flexibility to collaborate across Paris and Santa Clara timezones.

CORE COMPETENCIES

Deep Learning Training & Optimization | Reproducible Benchmarking Pipeline | Signal Processing | State-Space Models | Model Architecture Comparison | Python & PyTorch | Statistics & Machine Learning | Transformer & Autoencoder Architectures

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Constructed an end-to-end ML pipeline on **100K+ fiber-optic network records** (upload speed, latency, jitter) using Random Forest classification and K-Means clustering to assess network quality across thousands of nodes.
- Developed a reproducible clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with comparative results adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and cross-validated performance reporting.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, visualized through **Power BI** dashboards for non-technical decision-makers.

PROJECTS

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80 Unsupervised Fault Detection

- Designed an unsupervised fault-detection system for slide-rail machinery using a **transformer autoencoder on Mel spectrograms** with SpecAugment data augmentation, achieving AUC 0.80 with no labeled anomaly data - directly aligned with signal processing and modern architecture requirements.
- Benchmarked autoencoder reconstruction-error thresholds against baseline statistical methods, producing comparative results across architectures for predictive maintenance using **PyTorch and deep learning optimization** best practices.

PyTorch · Transformer Autoencoder · Mel Spectrograms · SpecAugment · Scikit-learn

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress June 2026 · Top 10% · \$106K+ Prize Pool

- Achieved **top 10% public leaderboard score (0.80/1.0)** in a \$106K+ Kaggle competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples using SFTTrainer and completion-only loss.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict sequence - including monkey-patched CUDA allocators, stubbed Mamba/Triton imports, and PyTorch fallback for rmsnorm - and conducted 20+ single-variable ablations confirming packing, synthetic data, and adapter-size effects on OOD generalization.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Mamba · Triton · TRL

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · Ranked 1st in EURECOM Class (0.791 AUC)

- Ranked **1st in EURECOM class private leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head across 250K images from 25+ generator types, using dual learning rates and FP16 mixed precision on T4 GPU; also submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Validated that **1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC)** via controlled ablation, and applied a 3-model ensemble weighted by validation AUC with 10-view test-time augmentation to improve out-of-distribution robustness.

PyTorch · OpenCLIP · ViT · AdamW · FP16 Mixed Precision · Kaggle

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Signal Processing, Computer Vision, NLP, Cloud Computing, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Deep Learning: PyTorch | Scikit-learn | Hugging Face Transformers | TRL / PEFT / Unsloth | TensorFlow | vLLM | OpenCLIP

Architectures & Techniques: Transformers | State-Space Models (Mamba) | Autoencoders | LoRA / QLoRA | CNN / DenseNet / ViT | Ensemble Methods | Feature Engineering

Programming, MLOps & Data: Python (expert) | SQL | Bash | Git / Linux | FP16/BF16 Mixed Precision | CUDA Debugging | Power BI | Pandas / NumPy